第二章 静电场

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基本概念

1. 静电场的源：电荷　点电荷间的伏拉克表示法 $\int_V \delta(r-r')dV=\begin{cases}0 & r'\text{不在}V\text{内}\\ 1 & r'\text{在}V\text{内}\end{cases}$

电荷密度与分布　体：$dq=\rho dV$，$\rho(r)=\lim_{\Delta V\to 0}\frac{\Delta q}{\Delta V}$

面：$dq=\rho dS$，$\rho_S(r)=\lim_{\Delta S\to 0}\frac{\Delta q}{\Delta S}$

线：$dq=\rho dl$　$\rho_l(r)=\lim_{\Delta l\to 0}\frac{\Delta q}{\Delta l}$

2. 库仑定律：$F_{21}=\frac{q_1q_2}{4\pi\varepsilon_0}\cdot\frac{e_{12}}{R^2}$　$F_{21}=-F_{12}$

3. 电场强度：E等于单位正电荷所受电场力

$E(x,y,z)=\lim_{q_t\to 0}\frac{F(x,y,z)}{q_t}$　V/m (N/C)

(a)单个点电荷产生的电场强度：$E=\frac{F}{q}=\frac{q}{4\pi\varepsilon_0R^2}\vec{e}_R$　V/m　或：$E(r)=\frac{q}{4\pi\varepsilon_0|r-r'|^2}\cdot\frac{r-r'}{|r-r'|}=\frac{q}{4\pi\varepsilon_0|r-r'|^3}(r-r')$

(b) n个点电荷产生的场强叠加：$E(r)=\frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{q_i}{R_i^2}\vec{e}_i=\frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{q_i(r-r_i')}{|r-r_i'|^3}$

(c)连续分布电荷产生电场强度：$dE=\frac{dq}{4\pi\varepsilon_0R^2}\vec{e}_R$　$dq=\rho dV/\rho_S dS/\rho_l dl$

体电荷分布：$E=\frac{1}{4\pi\varepsilon_0}\int_{V'}\frac{\rho dV'}{R^2}\vec{e}_R$　$\varphi=\frac{1}{4\pi\varepsilon_0}\int_{V'}\frac{\rho dV'}{R}$

面电荷分布 $E=\frac{1}{4\pi\varepsilon_0}\int_{S'}\frac{\rho_S dS'}{R^2}\vec{e}_R$　$\varphi=\frac{1}{4\pi\varepsilon_0}\int_{S'}\rho_S d$

线电荷分布 $E=\frac{1}{4\pi\varepsilon_0}\int_{l'}\frac{\rho_l dl'}{R^2}\vec{e}_R$　$\varphi=\frac{1}{4\pi\varepsilon_0}\int_{l'}\frac{\rho_l dl'}{R}$

4. 极化强度 $P=\lim_{\Delta V\to 0}\frac{\sum p}{\Delta V}$　C/m^2 电偶极矩体密度　在各向同性、线性、均匀介质中：$\vec{P}=\chi_e\varepsilon_0\vec{E}$　χ_e—电介质极化率

各向同性媒质：媒质特性不随电场方向改变，$D=\varepsilon_0E+P=\varepsilon_0E+\chi_e\varepsilon_0E=\varepsilon_r\varepsilon_0E=\varepsilon E$

线性媒质：媒质参数不随电场值而变化

均匀媒质：媒质参数不随空间坐标而变化　$\varepsilon_r=1+\chi_e$　$\varepsilon=\varepsilon_0\varepsilon_r$　$D=\varepsilon E$

$\varphi=\frac{q}{4\pi\varepsilon_0R}$ 为单个点电荷产生的电位

极化强度为电偶极矩体密度，单个电偶极子产生的电位：$\varphi=\frac{q\ell\cos\theta}{4\pi\varepsilon_0r^2}=\frac{1}{4\pi\varepsilon_0}\frac{\vec{P}\cdot\vec{e}_r}{r^2}$

偶极矩与极化强度 $P=\lim_{\Delta V\to 0}\frac{\sum p}{\Delta V}\Rightarrow$ V内电偶极子产生的电位 $\varphi=\frac{1}{4\pi\varepsilon_0}\int_{V'}\frac{P(r')\cdot(r-r')}{|r-r'|^3}dV'$

矢量恒等式：$\nabla\cdot(uF)=\nabla\cdot(uF)-u\nabla\cdot F$

$\varphi=\frac{-1}{4\pi\varepsilon_0}\int_V\frac{\nabla'\cdot P(r')}{R}dV'+\frac{1}{4\pi\varepsilon_0}\int_{V'}\nabla'\frac{P(r')}{R}dV'$

$=\frac{-1}{4\pi\varepsilon_0}\int_{V'}\frac{\nabla'\cdot P(r')}{R}dV'+\frac{1}{4\pi\varepsilon_0}\oint_{S'}\frac{P(r')\vec{e}_n}{R}d\vec{S}'$

$\varphi=\frac{1}{4\pi\varepsilon_0}\int_{V'}\frac{P(r')\cdot\vec{e}_R}{R^2}dV'$

$\because\frac{e_R}{R^2}=\nabla'\frac{1}{R}=-\nabla\frac{1}{R}$

$\therefore\varphi=\frac{1}{4\pi\varepsilon_0}\int_{V'}P(r')\cdot\nabla\frac{1}{R}dV'$

令 $\rho_P=-\nabla\cdot P$ 极化电荷体密度，极化电荷面密度

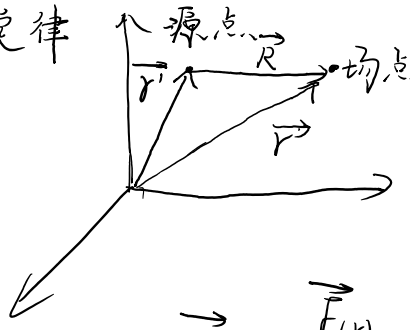
$\varphi(r)=\frac{1}{4\pi\varepsilon_0}\int_{V'}\frac{\rho_P(r')}{R}dV'+\frac{1}{4\pi\varepsilon_0}\oint_{S'}\frac{G_P(r')}{R}dS'$

根据电荷守恒定律，极化电荷的总和为零 $\int_{V'}-\nabla\cdot PdV'+\oint_{S'}\vec{P}\cdot\vec{e}_n\,d\vec{S}'=0$

电介质均匀极化后极化电荷体密度为 $\rho_P=0$

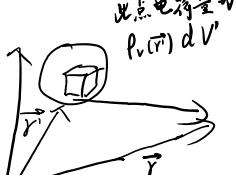
§2.1 库仑定律及电场强度

1. 库仑定律



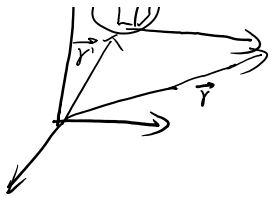
$\vec{R}=\vec{r}-\vec{r}'$

$\vec{F}(\vec{r})=\frac{q_1q_2}{4\pi\varepsilon_0}\frac{\vec{R}}{|R|^3}=\frac{q'q}{4\pi\varepsilon_0}\frac{\vec{r}-\vec{r}'}{|\vec{r}-\vec{r}'|^3}$



2. 电场强度：$\vec{E}(r)=\frac{\vec{F}(r)}{q'}$　$\vec{E}(\vec{r})=\frac{q'}{4\pi\varepsilon_0}\frac{\vec{r}-\vec{r}'}{|\vec{r}-\vec{r}'|^3}$

2. 电场强度：$\vec{E}(r)=\frac{\vec{F}(r)}{q}$　　$\vec{E}(\vec{r})=\frac{q'}{4\pi\varepsilon_0}\frac{\vec{r}-\vec{r}'}{|\vec{r}-\vec{r}'|^3}$



3. 分布电荷　电荷密度 ρ

① 体分布 ρ_v　$E(\vec{r})=\int_V \frac{\rho_v(\vec{r}')dV'}{4\pi\varepsilon_0}\frac{\vec{r}-\vec{r}'}{|\vec{r}-\vec{r}'|^2}dV'$

② 面分布 ρ_s　$E(\vec{r})=\int_V \frac{\rho_v(\vec{r}')dV'}{4\pi\varepsilon_0}\frac{\vec{r}-\vec{r}'}{|\vec{r}-\vec{r}'|^2}dV'$

③ 线分布 ρ_l　$E(\vec{r})=\int_l \frac{\rho_l(\vec{r}')dl'}{4\pi\varepsilon_0}\frac{\vec{r}-\vec{r}'}{|\vec{r}-\vec{r}'|^2}dl'$

§2.2 静电场中的高斯定理

1. 高斯定理：闭合曲面电通量等于其包围的电荷量

公式：积分形式　$\oint_S \vec{E}\cdot d\vec{S}=\int_V \nabla\vec{E}\cdot dV$　　$\oint_S \vec{E}\cdot d\vec{S}=\frac{Q}{\varepsilon_0}$

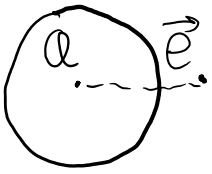
$Q=\int_V \rho(r)dV=\int_V \nabla\cdot\vec{D}\cdot dV$, $\rho=\nabla\cdot\vec{D}$

微分形式　$\nabla\cdot\vec{E}=\frac{\rho(r)}{\varepsilon_0}$

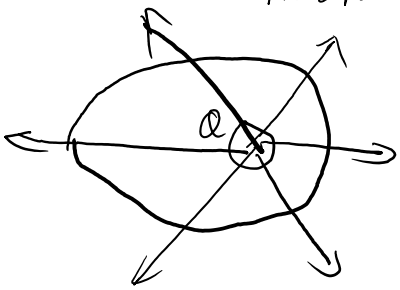
由库仑定律得到高斯定理　$\oint \vec{E}\cdot d\vec{S}=\oint_S |E|dS=\oint_S \frac{Q}{4\pi\varepsilon_0}\frac{1}{R^2}dS$

$=\frac{Q}{4\pi\varepsilon_0 R^2}\cdot 4\pi R^2=\frac{Q}{\varepsilon_0}$

① 球面+点电荷

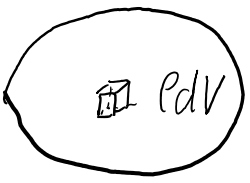


② 闭合曲面内+点电荷



③ 闭合曲面内+分布电荷

$\int_V \frac{\rho dV}{\varepsilon_0}=\frac{Q}{\varepsilon_0}$



④ 外电荷



$\Phi_{外}=0$

§2.3 电位

1. 重要理论：$\frac{\vec{r}-\vec{r}'}{|\vec{r}-\vec{r}'|^3}=-\nabla\frac{1}{|\vec{r}-\vec{r}'|}$

证明：$\vec{r}=x\vec{e}_x+y\vec{e}_y+z\vec{e}_z$，$r'=x'\vec{e}_x+y'\vec{e}_y+z'\vec{e}_z$

$\vec{R}=\vec{r}-\vec{r}'=(x-x')\vec{e}_x+(y-y')\vec{e}_y+(z-z')\vec{e}_z$

$|R|=|\vec{r}-\vec{r}'|=\sqrt{(x-x')^2+(y-y')^2+(z-z')^2}$

$$-\nabla\frac{1}{|\vec{r}-\vec{r}'|}=-\nabla\frac{1}{|R|}=\frac{\vec{R}}{|R|^3}=-\frac{1}{R^2}\frac{x-x'}{|R|}\vec{e}_x-\frac{1}{|R|^2}\frac{y-y'}{|R|}\vec{e}_y-\frac{1}{|R|^2}\frac{z-z'}{|R|}\vec{e}_z$$

$$=-\frac{1}{|R|^3}\left[(x-x')e_x-(y-y')e_y-(z-z')e_z\right]$$

综上：$\frac{\vec{r}-\vec{r}'}{|\vec{r}-\vec{r}'|}=-\nabla\frac{1}{|\vec{r}-\vec{r}'|}$

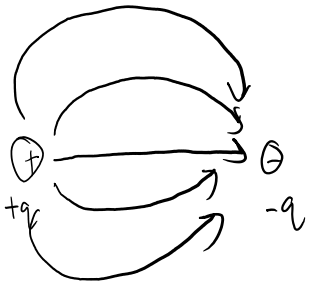
电位：$\vec{E}(r)=\int_{V'}\frac{\rho(r')}{4\pi\varepsilon_0}\frac{\vec{r}-\vec{r}'}{|\vec{r}-\vec{r}'|^3}dV'=\int_{V'}\frac{\rho(r')}{4\pi\varepsilon_0}\left(-\nabla\frac{1}{|\vec{r}-\vec{r}'|}\right)dV'$

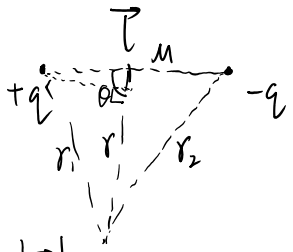
$$=-\nabla\left[\int_{V'}\frac{\rho(r')}{4\pi\varepsilon_0}\frac{1}{|\vec{r}-\vec{r}'|}dV'\right]$$

$$\varphi(r)=\int_{V'}\frac{\rho(r)}{4\pi\varepsilon_0}\frac{1}{\vec{r}-\vec{r}'}dV'+C$$

$\vec{E}(r)=-\nabla\varphi$

§2.4 电偶极子





$r\gg|l|$ 电偶极矩：$\vec{p}=q_1\cdot\vec{l}$，$|\vec{l}|$的数量级为10^{-9}

$r_1=r-\frac{|l|}{2}\cos\theta$，$r_2=r+\frac{|l|}{2}\cos\theta$

$$\varphi=\frac{q}{4\pi\varepsilon_0}\left(\frac{1}{r_1}-\frac{1}{r_2}\right)=\frac{q_1}{4\pi\varepsilon_0}\frac{|l|\cos\theta\, r}{r^2\cdot r}=\frac{\vec{p}\cdot\vec{r}}{4\pi\varepsilon_0 r^3}$$

① 空间直角坐标系下：$r=\sqrt{x^2+y^2+z^2}$

$$\Phi(x,y,z)=\frac{p}{4\pi\varepsilon_0}\cdot\frac{z}{(x^2+y^2+z^2)^{\frac{3}{2}}}=\frac{p}{4\pi\varepsilon_0}z\,r^{-3}$$

$$E=-\nabla\Phi=-\left(\frac{\partial\Phi}{\partial x}\vec{e}_x+\frac{\partial\Phi}{\partial y}\vec{e}_y+\frac{\partial\Phi}{\partial z}\vec{e}_z\right)$$

对x求偏导：$\frac{\partial\Phi}{\partial x}=\frac{pz}{4\pi\varepsilon_0}\frac{\partial r^{-3}}{\partial x}=\frac{3pxz}{4\pi\varepsilon_0 r^5}$

对y求偏导：$\frac{\partial\Phi}{\partial y}=\frac{pz}{4\pi\varepsilon_0}\cdot\frac{\partial r^{-3}}{\partial y}=\frac{3pyz}{4\pi\varepsilon_0 r^5}$

对z求偏导：$-p\frac{\partial}{4\pi\varepsilon_0 r^3}=-\frac{p}{}\frac{r^2-3z^2}{}$

$\frac{\partial \Phi}{\partial x} = \frac{P}{4\pi\varepsilon_0} \frac{\partial}{\partial x} = \frac{}{4\pi\varepsilon_0 r^5}$ 对y求偏导：$\frac{\partial \Phi}{\partial y} = \frac{P}{4\pi\varepsilon_0} \cdot \frac{\partial}{\partial y} = \frac{3yz}{4\pi\varepsilon_0 r^5}$

对z求偏导：$\frac{\partial \Phi}{\partial z} = \frac{P}{4\pi\varepsilon_0} \frac{\partial \frac{Pz}{4\pi\varepsilon_0 r^3}}{\partial z} = \frac{P}{4\pi\varepsilon_0} \frac{r^2-3z^2}{r^5}$

综上：$\vec{E}(x,y,z) = \frac{P}{4\pi\varepsilon_0 r^5}[3xz\vec{e}_x + 3yz\vec{e}_y + (3z^2-r^2)\vec{e}_z]$

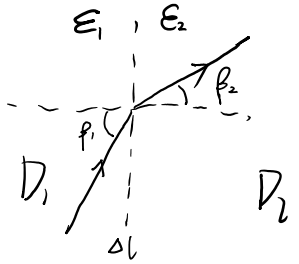
② 圆柱坐标系(ρ, ϕ, z)下，$\rho = \sqrt{x^2+y^2}$，z，ϕ（方位角）

$\nabla \Phi = \frac{\partial \Phi}{\partial \rho}\vec{e}_\rho + \frac{1}{\rho}\frac{\partial \Phi}{\partial \phi}\vec{e}_\rho + \frac{\partial \Phi}{\partial z}\vec{e}_z$

Φ与ϕ无关，因为：在圆柱坐标系下，一对电偶极子形成的电场电势由ρ、z的决定，形成等电势的环。对于每个ρ，z，都有一个等势环，不同的方位角与等势环上的点一一对应，球坐标系同理

分界面上的衔接条件

① 电通量密度D



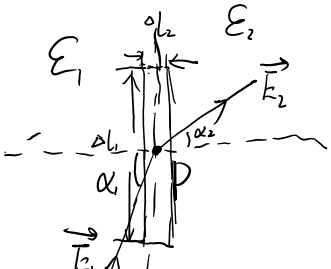
$\oint_S \vec{D}\cdot d\vec{S} = q$ (Gauss Theorem)

$-D_{1n}\Delta S + D_{2n}\Delta S = \rho_S \Delta S$

$D_{2n} - D_{1n} = \rho_S$，D的法向分量不连续，即界面上有自由电荷

当$\rho_S = 0$时，$D_{1n} = D_{2n}$ D的法向分量连续，界面上无自由电荷

② 切线上



$\oint_L \vec{E}\cdot d\vec{l} = 0$

则：$E_{1t}\Delta l_1 - E_{2t}\Delta l_1 = 0$

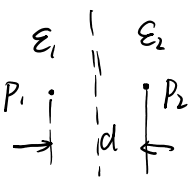
$E_{2t} = E_{1t} \Rightarrow$ E的切向分量连续

得到折射定理：当交界面$\rho_S = 0$时

$\begin{cases} D_{1n} = D_{2n} \Rightarrow \varepsilon_1 E_1 \cos\alpha_1 = \varepsilon_2 E_2 \cos\alpha_2 \\ E_{1t} = E_{2t} \Rightarrow E_1 \sin\alpha_1 = E_2 \sin\alpha_2 \end{cases}$

$\frac{\tan\alpha_1}{\tan\alpha_2} = \frac{\varepsilon_1}{\varepsilon_2}$ 为折射定律

③ Φ的衔接条件 $\Phi_1 - \Phi_2 = \lim_{d\to 0}\int_0^d E\,dl = 0$，$\Phi_1 = \Phi_2$，电位连续

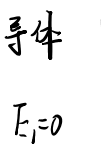


$\because D_{2n} - D_{1n} = \rho_S$

$\therefore D_{1n} = \varepsilon_1 E_{1n} = -\varepsilon_1 \frac{\partial \Phi_1}{\partial n}$，$D_{2n} = \varepsilon_2 E_{2n} = -\varepsilon_2 \frac{\partial \Phi_2}{\partial n}$

综上：$\varepsilon_1 \frac{\partial \Phi_1}{\partial n} - \varepsilon_2 \frac{\partial \Phi_2}{\partial n} = \rho_S$ 电位的法向导数不连续，单位向量$\vec{n}$由介质1指向2

④ 导体与电介质分界面上的衔接条件



$D_{2n} - D_{1n} = \rho_S$ $E_{1t} = E_{2t}$，$\Phi_1 = \Phi_2$

$\varepsilon_1 \frac{\partial \Phi_1}{\partial n} - \varepsilon_2 \frac{\partial \Phi_2}{\partial n} = \rho_S$ 导体中$E = 0$，

分界面只要不是良导体，分界面就没有自由电荷密度

有自由电荷流过，分界面就有自由电荷密度

2.3 电容

定义：$C=\frac{Q}{U}$ 单位 F

电容只与两导体的几何尺寸、相互位置及周围介质有关.

工程上电容器分为电力电容、电子电容

$Q\rightarrow E\rightarrow U=\int_L \vec{E}\cdot d\vec{l}\rightarrow C=\frac{Q}{U}$